

Doppler ultrasonography in suspected intrauterine growth retardation: a randomized clinical trial

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ABSTRACT

A randomized clinical trial was performed to test the hypothesis that if suspected intrauterine growth retardation (IUGR) is associated with normal umbilical artery Doppler ultrasound findings, hospitalization can safely be avoided.

One hundred and fifty women with singleton pregnancies and suspected IUGR were randomized between an intervention (n = 74) and a control group (n = 76). In the intervention group, clinicians were strongly requested not to hospitalize for suspected IUGR if the Doppler findings were normal. In the control group, the Doppler results were not revealed and the participants received the standard management for suspected IUGR. Endpoints of the trial were: costs in terms of hospitalization, perinatal outcome, neurological development, and postnatal growth.

Duration of hospitalization was significantly shorter in the intervention group than in the control group. Contrary to expectations, the hospitalization rate during pregnancy in the intervention group was not below that of the control group. This negative finding was partly due to the admission of patients in the intervention group despite their normal Doppler results. Moreover, the trial might have induced a more critical attitude towards hospitalization in suspected IUGR, decreasing admission in the control group. No clear differences were found in perinatal outcome, neurological development, or postnatal growth.

The results suggest that normal umbilical artery Doppler findings in suspected cases of IUGR justify outpatient management.

INTRODUCTION

Intrauterine growth retardation (IUGR) is the condition whereby the fetus cannot fulfil its genetic growth potential. IUGR is associated with an increased risk of perinatal

morbidity and mortality¹. If IUGR is suspected during pregnancy, the mother is often hospitalized for fetal surveillance.

IUGR is conventionally defined by certain thresholds of birth weight, corrected for the gestational age. In obstetric decision making, this parameter is estimated during pregnancy by fundal height measurements or ultrasonic fetal biometry. Such thresholds, however, define those fetuses or neonates who are small for gestational age (SGA). That SGA is by no means synonymous with IUGR is illustrated in Figure 1, where the absence or presence of SGA is compared with the absence or presence of IUGR. The neonates on the left side of each square symbolize the genetic growth potential and those on the right the actual size reached at birth. If there are no detrimental influences in the intrauterine environment, the neonate at birth is identical to its genetic counterpart (IUGR not present). The figure shows that the conventional thresholds lead to a detectable true-positive diagnosis of IUGR in the lower right quadrant. Simultaneously, a 'false' positive diagnosis can be obtained due to genetic smallness of the fetus or a restricted predictive value of the test.

Compromized utero-placental circulation is thought to play an important role in the development of IUGR. Some authors have suggested that suspected IUGR is a largely benign condition if it is associated with normal blood velocity profiles in the umbilical artery^{2–5}. If this positive relationship between normal umbilical artery flow and fetal wellbeing is true, normal Doppler results are expected in cases of genetic or false-positive smallness. Unnecessary and costly clinical management can be avoided in these low-risk cases. Distinguishing the false-positive cases from actual IUGR cases using Doppler ultrasonography is, therefore, attractive from a cost point of view.

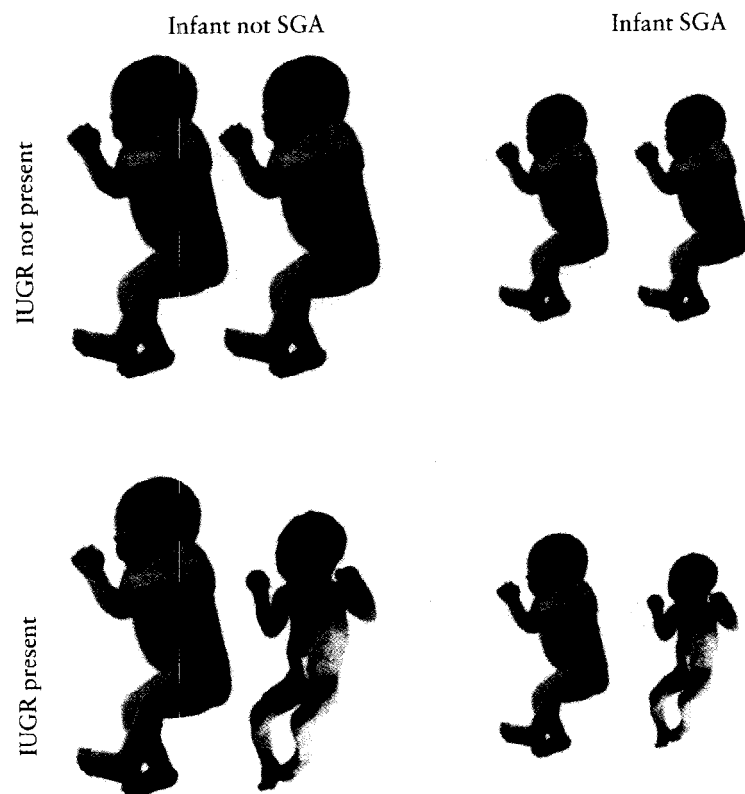


Figure 1 Small-for-gestational age (SGA) and intrauterine growth retardation (IUGR). The neonates on the left side of each square symbolize the genetic growth potential and those on the right the actual size reached at birth

The objective of the randomized clinical trial reported in this paper was to test the hypothesis that the use of Doppler ultrasonography as a reassurance test in suspected cases of IUGR could reduce costs, without this having a detrimental effect on perinatal outcome.

METHODS

The study was performed at the University Hospital Maastricht in the Netherlands. At the time of the study, Doppler ultrasonography was not routinely available for clinical management. Therefore, withholding Doppler information from patients in the control group did not constitute an ethical problem. The study protocol was approved by the Hospital Ethics Committee. The number of eligible patients was estimated at 16% of the singleton pregnancies at the outpatient clinic on the basis of the birth centile distribution in previous years. As the actual accrual rate is generally half the estimated rate⁶, and taking into account the fact that the accrual period of a study should be limited to approximately 2.5 years, an estimated 150 participants could be randomized. This logistically feasible study group was used to provide some insight into the above hypothesis.

Inclusion started in September 1989 and the follow-up was completed in March 1993. The sole inclusion criterion was a clinically suspected IUGR of ≥ 2 weeks, diagnosed by

fundal height measurements at the outpatient clinic. Exclusion criteria were multiple pregnancy, uncertain gestational age, non-Caucasian origin, and maternal or fetal conditions requiring immediate hospitalization or intervention. Gestational age was determined by early ultrasonic fetal biometry, which was generally available because the hospital provides ultrasound dating as a basic regional service. Participants were randomly allocated to an intervention and a control group. Before randomization, participants were stratified according to the gestational age at randomization (≤ 32 weeks or > 32 weeks) and smoking habits (non-smoker or smoker regardless of the number of cigarettes smoked.) The stratification criteria led to four randomization lists which were prepared in advance from a published table of random numbers. A randomization number was requested over the telephone from an independent person not involved in patient management.

Doppler ultrasound examinations were performed weekly until delivery in both the intervention and the control groups. All examinations were performed by one examiner (SJN). Examinations were scheduled directly before the patient's visit to the outpatient clinic. A loop of umbilical cord was identified by ultrasound scanning (Pie Medical 1120 scanner, Pie Medical, Maastricht, The Netherlands). The angle between the ultrasound beam and the direction of flow was minimized and the sample volume (approximately 5×5 mm) placed so that it enclosed the

whole vessel. The ultrasound transducer was linked to a 2-MHz pulsed Doppler device. In Doppler mode, the quality of the signal was optimized by ear and by on-line display of the signal. Registrations were only accepted if the beat-to-beat variation was low, indicating fetal quiescence without fetal breathing and excessive fetal movements. Low-frequency vessel wall movements were eliminated by a 50-Hz high-pass filter. A digital two-channel sound recording (Sony Super VHS, Japan) was made of the quadrature audio information of ten or more consecutive velocity waveforms for off-line analysis.

The pulsatility index (PI) was computed semi-automatically by tracing the maximum frequency outline of the Doppler shift waveforms with a lightpen on the monitor of the spectrum analyzer (Doptek, Vingmed, Norway). Doppler results were not corrected for fetal heart rate, as no measurements were made outside the physiological range of 120–160 beats per minute⁷. The final PI result was calculated from six consecutive waveforms to provide a representative value^{8,9}. This result was compared to the reference values for the gestational age previously established in a longitudinal study of umbilical artery flow in uneventful singleton pregnancies at our outpatient clinic¹⁰. These reference values correspond well with other published values^{11–13}. The PI results were divided into three color-coded groups: green, orange and red. Green indicated a PI value within the normal range, defined as a PI below the mean plus one standard deviation (SD) for the gestational age ($PI < \text{mean} + SD$). Red indicated a raised PI result, defined as a PI above the mean plus two SD ($PI > \text{mean} + 2SD$) for the gestational age. This category also included zero end-diastolic or reversed flow. A value between one and two standard deviations above the mean ($\text{mean} + SD \leq PI \leq \text{mean} + 2SD$) for the gestational age was defined as of indecisive clinical value and conveyed by the color orange. For analysis, the ‘worst’ PI result obtained during the study period was used, as that result could lead to hospitalization.

In the intervention group, the Doppler ultrasound results were conveyed to the clinician or midwife in attendance by a colored label in the patient’s notes. The Doppler results were linked to a predefined hospitalization policy. In cases of a ‘green’ Doppler result, the clinician in charge was strongly requested *not to hospitalize* the participant for IUGR during pregnancy. In cases of a ‘red’ Doppler result, the clinical management was left to the clinician in charge. In cases of an ‘orange’ Doppler result, the examination was repeated within 3–4 days. The same consequences were applied to the results of these extra examinations. Meanwhile, the policy was not to hospitalize for IUGR. In summary, the intervention consisted of maintaining outpatient management while the Doppler was in the normal range, in a setting whereby hospitalization was the management of choice if significant IUGR was suspected.

In the control group, the PIs were not calculated until after delivery and the results were concealed from the clinician in charge. All participating clinicians were well aware that Doppler results in the control group would not be revealed, whatever the result. Participants in this group

received the standard clinical management for suspected IUGR.

The endpoints of the randomized clinical trial were: effect on costs in terms of hospitalization, perinatal outcome, neurological development, and postnatal catch-up growth. For the cost analysis, medical procedures of all participants were tallied from the database of the hospital’s financial administration. As hospitalization formed the major component of cost, this was analyzed primarily. Parameters of obstetric management and perinatal outcome corresponded to those traditionally presented in published Doppler trials and included the onset and mode of delivery, birth weight, and gestational age at birth.

Possible detrimental effects of the Doppler-based intervention were assessed by repeated examinations of the neurological development of the infants by an experienced child neurologist who was unaware of the Doppler results. Between 3 and 8 days after term delivery, the neonates were examined according to the method developed by Prechtl¹⁴. The full Prechtl examination was performed and the results were noted on the published summary form. Age-specific neurological examinations¹⁵ were performed at the corrected ages of 6 and 9 months ± 2 weeks. Preterm infants were examined within 8 days of reaching term (‘gestational’ age of 37 weeks or 259 days) and further examination dates were corrected for the gestational age in accordance with the literature^{16–18}. Infants were examined on the neonatal ward or in the outpatient department in a warm room for approximately 20 minutes. The overall result of each examination was categorised as being normal, suspect, or abnormal.

In the absence of a definitive standard for IUGR it was hypothesized that the occurrence of postnatal catch-up growth might be used for this purpose. Repeated anthropometric measurements of weight, height, and head circumference were performed on all infants until the corrected age of 9 months. The weight of the fully undressed infant was determined in grams. Crown-to-heel length or height was measured with the aid of a horizontal measuring board. An assistant held the child’s head in the Frankfurter plane whereby the lower borders of the orbits were in the same plane as the external auditory meati. The infant’s legs were straightened with the ankles at right angles while the footboard was pulled firmly against the child’s heels. Height could be read to the nearest millimeter from the built-in digital counter. Head circumference was measured in millimeters with the tape placed over the most extending point of the occiput and just above the ridges of the eyebrows. Reference growth curves were based on a Dutch national neonatal growth study¹⁹. The measurements were corrected for age and sex by expressing them as a deviation from this reference, the so called standard deviation score (SDS)²⁰:

$$SDS = \frac{\text{measurement} - \text{mean for age}}{SD \text{ for age}}$$

On the basis of the literature, catch-up growth was defined as an upward shift in anthropometric parameters of ≥ 1 SDS in the first 6 months^{21,22}.

With regard to blinding, it was inherent in the nature of the trial that the clinicians in charge could not be blinded with regard to the group to which the participant was assigned. Neither was it possible to blind participants in the intervention group, as they could be asked to return earlier for a repeat examination if the Doppler result was indecisive (orange). However, the Doppler ultrasound results in the control group were concealed from clinicians in charge. The examiners performing the neurological and anthropometric follow-up were unaware of the Doppler results.

Data were collected on data forms and entered into a database. Statistical analysis was performed with SPSS-PC + SPSS Inc., Chicago, USA. Differences between the intervention and the control groups in hospitalization rate, neurological development and postnatal growth were tested by comparing the means with a *t*-test, and the percentages with a χ^2 -test with $\alpha = 0.05$ and $\beta = 0.1$. A Mann-Whitney test was used to test the difference in duration of hospitalization between the groups, as these durations were not normally distributed. The differences between the study groups are presented with the associated 95% confidence interval (CI).

RESULTS

Of the 161 women referred because of suspected IUGR of ≥ 2 weeks, 150 (93%) consented to join the study. The reasons for refusal were lack of time and plans to move out of the area. Of the 150 women, 74 participants were randomized to the intervention group and 76 to the control group. Table 1 shows the characteristics of the randomization groups at inclusion. Stratification and randomization led to comparable groups with regard to most parameters. By chance some parameters were not equally distributed over the intervention and the control groups, namely fetal sex (58.1% and 36.8% boys, respectively) and breech presentations (4.1% and 18.4%, respectively). In the analysis, the possible influence of the skewed distribution of sex was reduced by using sex-specific growth references. The number of hospital days solely attributable to breech

presentation was low (intervention group: no days during pregnancy, one neonatal hospital day, one maternal postnatal hospital day; control group: no days during pregnancy, five neonatal hospital days, five maternal postnatal hospital days). The influence on the analysis was therefore small and no form of correction was made.

Table 2 shows the Doppler ultrasound results and perinatal outcome in the intervention group and the control group. An average of 5.9 Doppler measurements were performed in the intervention group (range 1–15) and 5.7 in the control group (range 1–16). The distributions of the traditional parameters for perinatal outcome are comparable.

Four cases were excluded in both the intervention and control groups because of congenital defects. One intrauterine fetal death (IUFD) in the control group had multiple abnormalities. One neonatal death in the intervention group was due to a large ventricular septal defect. Three infants survived despite cardiac defects, one in the intervention group and two in the control group. One intervention group infant had spina bifida and hydrocephalus. In one control group infant, trisomy 21 was diagnosed. During follow-up, one intervention group infant developed hydrocephaly due to intracerebral abnormalities, necessitating the implantation of a drain. Such congenital defects can be associated with extended medical treatment, lower genetic growth potential or restricted postnatal growth. As it then becomes impossible to evaluate the possible effect of IUGR on costs, development and growth, these cases were excluded from further analysis. In conclusion, 70 infants could be analyzed for IUGR in the intervention group and 72 in the control group.

Normally-formed IUFD due to a severely deficient utero-placental circulation occurred once in the intervention group and twice in the control group. In all three cases, umbilical Doppler findings were abnormal. In the IUFD in the intervention group, the parents decided to abstain from obstetric intervention because of the prematurity and the low estimated fetal weight. Two intervention group infants experienced meningitis with permanent

Table 1 Characteristics of the study groups at inclusion

Characteristic	Intervention group (<i>n</i> = 74)	Control group (<i>n</i> = 76)
Smokers	38 (51%)	37 (49%)
GA $\leq$ 32 weeks	26 (35%)	30 (39%)
Maternal age at LMP (years)	29 $\pm$ 5 (19–42)	28 $\pm$ 5 (19–42)
Maternal height (cm)	167 $\pm$ 6 (150–181)	165 $\pm$ 7 (145–183)
Pre-pregnancy weight (kg)	58 $\pm$ 8 (42–80)	56 $\pm$ 9 (40–80)
Maternal weight gain (kg)	11 $\pm$ 4 (2–22)	11 $\pm$ 4 (2–22)
Paternal height (cm)	178 $\pm$ 7 (160–193)	178 $\pm$ 8 (160–210)
Paternal weight (kg)	78 $\pm$ 11 (52–110)	78 $\pm$ 11 (55–104)
Primipara	34 (45.9%)*	43 (56.6%)
GA at diagnosis of IUGR (days)	230 (137–269)	225 (133–269)
GA at inclusion (days)	233 (137–271)	233 (159–283)

Results are shown as number and percentage, or mean $\pm$ standard deviation and range. Parity is shown as the number and percentage of primipara, and gestational age as median and range. LMP, last menstrual period; GA, gestational age; **p* = 0.19 compared with control group (χ^2 test)

Table 2 Obstetric management and perinatal outcome

Perinatal outcome	Intervention group (n = 74)	Control group (n = 76)
'Green' Doppler result	42 (56.7%)	45 (59.2%)
'Orange' Doppler result	9 (12.2%)	7 (9.2%)
'Red' Doppler result	15 (20.3%)	18 (23.7%)
'Red' Doppler: zero/reversed flow	8 (10.8%)	6 (7.9%)
Spontaneous onset of delivery	51 (68.9%)	51 (67.1%)
Induction of labor/elective CS*	21 (28.4%)	21 (26.7%)
Induction of labor not for IUGR**	2 (2.7%)	6 (7.9%)
Spontaneous delivery	63 (85.1%)	57 (75.0%)
Elective CS	6 (8.1%)	4 (5.3%)
Emergency CS	2 (2.7%)	7 (9.2%)
Forcipal extraction	3 (4.1%)	7 (9.2%)
Vacuum extraction	0 (0)	1 (1.3%)
Deviant cardiotocograph†	30 (40.5%)	27 (35.5%)
Breech presentation	3 (4.1%)* **	14 (18.4%)
Preterm birth	13 (17.6%)	13 (17.1%)
IUFD (due to defects)‡	1(0) (1.4%)	3(1) (3.9%)
Neonatal death (due to defects)‡	1(1) (1.4%)	0(0) (0%)
Male infant	43 (58.1%)	28 (36.8%)
Female infant	31 (41.9%)* **	48 (63.2%)
Birth weight ≤ 2500 g	29 (39.2%)	24 (31.6%)
GA at birth (days)	273 (200–299)	275 (184–295)
Birth weight (g)	2737 (640–3975)	2695 (480–3790)
First test to delivery interval	38 (0–135)	36 (4–110)

Results are shown as number and percentage, or as median and range. *, includes prostaglandin or oxytocin induction of labor and elective Cesarean section for IUGR, does not include induction of labor for IUFD or chemical stimulation because of ruptured membranes without contractions; **, indications in intervention group were intrauterine fetal death, Strumpell's syndrome; indications in control group were three IUFDs, post-date pregnancy, bleeding, ulcerative colitis; †late deceleration or bradycardia on the cardiotocograph; ‡includes chromosomal and/or congenital defects; CS, Cesarean section; IUGR, intrauterine growth retardation; IUFD, intrauterine fetal death; ***, $p = 0.01$ compared with control group (χ^2 test)

Table 3 Duration of hospitalization (in days) for suspected intrauterine growth retardation (IUGR)

Hospitalization for suspected IUGR	Intervention group (n = 70)		Control group (n = 72)		Mann-Whitney test p-value	Difference between medians	95% CI of the difference
	Total	Median	Total	Median			
During pregnancy	343	7.5	421	18.5	0.02*	-5.0	-13 to -1
Without protocol deviations	183	8.5	421	18.5	0.04*	-5.0	-13 to 0
Neonatal	588	9.0	636	10.0	1.00	0	-5 to 4
Postnatal	188	3.0	169	4.0	0.98	0	-1 to 1

*Significantly different

neurological damage just before their neurological examination at the age of 6 months. Data before that point in time were included in the analysis.

Procedures leading to costs associated with suspected IUGR included outpatient visits, ultrasound scans, cardiotocographic monitoring and hospitalization. Hospital admissions made up the major part of the total costs. Hospital admission during pregnancy was significantly shorter in the intervention group than in the control group (Table 3, $p = 0.02$). This was primarily because participants in the intervention group were managed longer as outpatients than those in the control group. This is re-

flected by a longer period between inclusion and hospital admission and by a higher gestational age at admission in the intervention group compared to the control group. Average neonatal hospital days were not significantly different in the groups ($p = 1.0$); neither were maternal postnatal hospital days.

Contrary to expectations, the proportion of women hospitalized during pregnancy was not significantly different in the intervention and the control groups (Table 4, $p = 0.24$). The neonatal hospitalization rate was similar in the two groups, as expected because hospitalization was theoretically prevented in low-risk cases only. Hospital

pediatric care was required in 53.6% (37/69) of the live-born infants in the intervention group and in 52.9% (37/70) in the control group.

Procedures such as outpatient visits, cardiotocographs and ultrasound scans, were tallied retrospectively from the hospital financial administration. Their occurrence was comparable in the intervention and in the control groups.

The occurrences of abnormal neurological examination results after delivery and at the ages of 6 and 9 months are presented in Table 5. The results of the neurological examinations did not differ significantly between the two study groups at any of the three examinations.

Anthropometric parameters of live-born infants were measured an average of ten times in the intervention group (range 3–16) and ten times in the control group (range 4–14). The anthropometric values of weight, height, and head circumference of each infant were plotted against age. On visual scrutiny, the resulting individual growth curves were found to progress smoothly without outliers or sharp bends. The longitudinal postnatal growth process was therefore summarized by two measurement points. The standard deviation scores of the measurements were calculated by using the formulas for the mean and standard deviations of the constructed reference curves. The differ-

ence in SDS between the anthropometric measurements after delivery and the measurements around the 6th month after delivery were used to establish the presence of catch-up growth. The results are shown in Table 6. The occurrence of catch-up growth in weight did not differ significantly between the two groups; neither did the occurrence of catch-up in height or head circumference.

DISCUSSION

The objective of this randomized clinical trial was to test the hypothesis that the use of Doppler ultrasonography as a reassurance test in suspected cases of IUGR could, by identifying the low risk cases, reduce hospitalization in comparison to the 'standard' management, without this having a detrimental effect on perinatal outcome.

The intervention was expected to reduce maternal hospitalization during pregnancy, by preventing the admission of cases of genetic fetal smallness associated with normal Doppler results. Although the trial led to a significant reduction in the number of hospital days in the intervention group in comparison to that of the control group, in accordance with the protocol, the hospitalization rate

Table 4 Hospitalization rates for suspected intrauterine growth retardation (IUGR)

	Intervention group (<i>n</i> = 70, infants 69)		Control group (<i>n</i> = 72, infants 70)		<i>p</i> Value*	Difference (%)	95% CI of the difference
	<i>n</i>	%	<i>n</i>	%			
During pregnancy	32	45.7	26	36.1	0.24	9.6	−6.2 to 25.7
Without protocol deviations	18	25.7	26	36.1	0.18	−10.4	−25.5 to 4.7
Neonatal	37	53.6	37	52.9	0.93	0.7	−15.8 to 17.4
Postnatal	36	51.4	37	51.4	1.00	0	−16.4 to 16.5

*Significance calculated using χ^2 test

Table 5 Neurological development

Abnormal neurological development	Intervention group (infants, <i>n</i> = 70)		Control group (infants, <i>n</i> = 72)		<i>p</i> Value*	Difference (%)	95% CI of the difference
	<i>n</i>	%	<i>n</i>	%			
After delivery	6/67	9.0	11/66	16.7	0.18	−7.7	−19.0 to 3.6
6 months	21/64	32.8	22/67	32.8	1.00	0	−16.1 to 16.1
9 months	7/67	10.4	12/70	17.1	0.26	−6.7	−18.2 to 4.8

The study group size varies because appointments were missed for various reasons not associated with the trial. Two intervention group infants experienced meningitis with permanent neurological damage just before their neurological examination at the age of 6 months. Data before that point in time were included in the analysis. Normal IUFD associated with placental insufficiency were included as having an abnormal examination result. *Significance calculated using χ^2 test

Table 6 Catch-up growth in the first 6 months after delivery

Catch-up growth	Intervention group (infants, <i>n</i> = 69)		Control group (infants, <i>n</i> = 70)		<i>p</i> Value*	Difference (%)	95% CI of the difference
	<i>n</i>	%	<i>n</i>	%			
Weight	23	33.3	34	48.6	0.07	−15.2	−31.4 to 0.9
Height	15	21.7	25	35.7	0.07	−14.0	−28.8 to 0.9
Head circumference	21	30.4	24	34.3	0.63	−3.9	−19.4 to 11.7

*Significance measured using χ^2 test

during pregnancy was comparable in both study groups. This could lead to the conclusion that the trial has failed or that its basic assumptions were wrong. However, this raises several objections.

Closer examination of indications for admission in the intervention group revealed that 14 participants were admitted during pregnancy despite their normal Doppler results. Suspected growth retardation was the sole reason for hospitalization in half of these protocol deviations. These cases should therefore not have been admitted to hospital according to the trial's protocol. In the remaining cases, additional indications besides the suspected IUGR influenced the decision to hospitalize. These included one case of suspected rupture of the membranes, in which loss of amniotic fluid could not be demonstrated and the patient was allowed to go home. She went into labor spontaneously at a later date. In three cases, the obstetric history of a former intrauterine fetal death influenced admission. In individual cases, additional indications besides the suspected IUGR, such as hypertension, decreased fetal movements and abdominal pain were also of influence on admission. The indications for hospitalization of the 14 protocol deviations were retrospectively and independently scrutinized by two gynecologists. In the opinion of these reviewers, the decision to hospitalize was understandable from the point of view of the clinician in charge in two cases: the case with the ruptured membranes and one case of suspected IUGR with a previous IUFD. If the protocol had been strictly adhered to and the protocol deviations had not been hospitalized, the difference in hospitalization rate would still not have reached significance (Table 4, $p = 0.18$). The effect of these protocol deviations may have been less if the study protocol had been enforced upon the participating clinicians. By enforcement the maximal effect of the intervention might be obtained (efficacy). The results, however, may not be extrapolated to clinical practice because the same effect might not be reached under less strict conditions²³. Moreover, enforcement can even be counter-effective. By giving the intervention a strongly advisory character, the results are those which can be obtained in clinical practice (effectiveness). An impression of the efficacy was obtained by analyzing the results both with and without the protocol deviations in Tables 3 and 4.

The characteristics of the 'standard' management in suspected IUGR were assessed separately in a questionnaire to which 80% of the obstetricians in the university hospitals in the Netherlands responded²⁴. They estimated that IUGR was suspected in 12% of their outpatient population and that approximately 70% of these cases were hospitalized for IUGR during pregnancy. The hospitalization rates during pregnancy found in the present randomized trial were 45.7% in the intervention and 36.1% in the control groups. This suggests that the estimated reduction in the intervention group of admissions during pregnancy was achieved, but that the hospitalization rate was lower than expected in the conventionally managed control group. The implementation of the study protocol could possibly

have led to a more critical attitude towards hospitalization in cases of suspected IUGR. This might suggest that sizeable savings can be achieved by a more critical attitude alone, regardless of the application of Doppler ultrasound techniques.

Perinatal outcome was comparable in the two study groups, indicating that there was no evident negative effect of the Doppler-based intervention. This is supported by a recent meta-analysis of published randomized clinical trials which showed that the availability of umbilical artery Doppler ultrasonography significantly decreases perinatal mortality²⁵. However, not every perinatal death should automatically be interpreted as an adverse outcome in the evaluation of a test. Perinatal death due to chromosomal and/or congenital defects is an entity distinct from IUGR and can be associated with normal Doppler ultrasound results. Exclusion of cases with defects in the above meta-analysis did not significantly influence the positive findings. Another objection is illustrated by the IUFD due to severe IUGR in the intervention group. The high risk was adequately identified by Doppler ultrasonography but the parents decided after ample consideration to abstain from obstetric intervention. Therefore, the outcome, had obstetric intervention taken place, could not be assessed.

No significant differences were found between the intervention and the control groups in the occurrence of abnormal neurological development. This indicates that the Doppler ultrasound-based policy does not seem detrimental to short-term neurological development. Most of the infants regarded as 'suspect' at the examination after delivery were found to be 'normal' at subsequent examinations. It is noteworthy that in both groups a number of infants found to 'normal' at the Prechtl examination and at 9 months, were found to be 'suspect' at the corrected age of 6 months. This finding is corroborated by the literature where transient abnormal neuromotor findings were shown to have a maximal expression between 6 and 8 months corrected age²⁶. The long-term consequences of these data are not clear.

There was no significant difference between the intervention and the control groups in the occurrence of catch-up growth. The confidence intervals of the difference between the groups showed a trend towards a lesser occurrence of catch-up in the intervention group. In conclusion, if IUGR is retrospectively identified by the occurrence of catch-up growth, no detrimental effect of the study protocol on this aspect of outcome could be found.

The results of this randomized clinical trial suggest that Doppler ultrasound measurements can be applied as a reassurance test in clinical practice to support the risk assessment of fetuses suspected of IUGR. Outpatient care if the Doppler ultrasound findings are within the normal range is less costly, less stressful to the patient, socially more acceptable and in this trial not evidently detrimental to neonatal outcome. While there is no proven beneficial effect of hospital admission, outpatient care is the management of choice in suspected cases of IUGR associated with normal Doppler results.

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